Research V2 / Computation Tracing

PDF edition of preserved repository document | docs/research_v2/computation_tracing.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Actual computation tracing

The v2 diagnostic instruments the preserved learned numeric trajectory policy. It is a small trained NumPyMLP, not Qwen/GPT/Gemini inference. No coding-capability improvement is established by it. Its real checkpoint, configuration and input pairs are pinned in configs/research_v2/trace_diagnostic.json; checkpoint files are checksum-verified while streaming the complete preserved export.

Each JSON trace records exact model-facing input, numeric normalization and every feature vector passed to the real forward method; there is no text tokenizer in this backend. Actual logits, tensor/weight shapes, derived uncalibrated softmax, timings, deterministic working-memory changes, policy transitions, output, independent verification and execution failures remain explicit. The hidden activation summaries are recomputed from the observed input and checkpoint, not falsely described as stack-captured or causal explanations.

Structured entities/relationships, planning search, dreaming, feedback learning and recovery are not implemented in this numeric backend; the trace says so. Coding trajectories implement additional typed state fields but need separate prospective neural instrumentation. High masked success may be driven by deterministic singleton legal-action constraints. A trace cannot establish neural contribution or superiority: controlled interventions and independent scoring are required.

trace_diagnostic.py generates one machine-readable JSON and readable Markdown report per actual diagnostic execution. The observer-preservation test compares answer and trajectory against the original predictor; the intervention test confirms unchanged weights. No hidden target enters the model-facing view. Diagnostic pairs are not a held-out competitive benchmark. The next study must freeze novel evaluation tasks and all interventions before evaluation.